

ENGINEERING SPECIFICATION

Tony Stark-Inspired Spatial Interaction System

Canonical Technical Specification

Real-time visual interaction, wearable sensing, sensor fusion, intelligent interaction, and display-agnostic spatial content control.

Document Property	Value
Document status	Canonical engineering specification
Revision	1.0
Scope	Three core development phases
Primary Phase 1 platform	MacBook Air with Apple Silicon M5, 10-core CPU, 10-core GPU, 24 GB unified memory, built-in webcam and display
Architecture posture	Content-agnostic, display-agnostic where practical, modular, instrumented
Implementation status	Specification only; implementation choices remain open unless explicitly constrained

AUTHORITY: This document is the authoritative requirements baseline for future implementation and review. A future implementation is considered conformant only when it satisfies the applicable requirements, phase gates, interfaces, and validation obligations defined here.

ROADMAP NORMALIZATION: The project is defined here as exactly three core development phases. Earlier five-phase planning material is superseded for roadmap numbering. Future spatial-display work and broader end-to-end optimization remain compatibility and post-core scope rather than additional core phases.

Prepared for engineering implementation, experimentation, benchmarking, and portfolio review

1. Document Purpose and Scope

This specification defines the system-level technical contracts, requirements, constraints, interfaces, quality attributes, validation framework, and phase gates for the Tony Stark-Inspired Spatial Interaction System. It is intentionally implementation-neutral. It does not select a programming language, library, model architecture, sensor part, rendering framework, or algorithm unless that choice is explicitly established as a constraint.

1.1 Purpose

- Provide a stable architectural reference against which implementation work can be designed, reviewed, tested, and accepted.
- Prevent accidental coupling between sensing, perception, interaction semantics, content, and presentation technologies.
- Ensure performance, latency, jitter, reliability, and observability are treated as first-class engineering requirements rather than post-hoc optimization tasks.
- Provide a controlled progression from webcam-only interaction to multimodal sensing and intelligent interaction.
- Preserve future compatibility with non-headset spatial display technologies without making such technologies a dependency of the core system.

1.2 System Boundary

In Scope	Out of Scope / Not Assumed
Real-time hand-mediated interaction with arbitrary supported 3D content.	Dependence on fictional MCU technology or physically impossible sensing/display claims.
Visual sensing, wearable sensing, sensor fusion, temporal state estimation, gesture/intent interpretation, interaction semantics, rendering/display abstraction, and instrumentation.	AR headset ownership or headset tracking as a required prerequisite.
Quantitative benchmarking, reproducible validation, environmental robustness, and phase-level acceptance gates.	Premature procurement of Phase 2 hardware before Phase 1 exit criteria are satisfied.
Future compatibility contracts for additional cameras, wearable sensors, renderers, and display technologies.	A fixed single-object demonstration as the architectural endpoint.

1.3 What the System Is

A real-time spatial interaction platform that converts human hand activity into structured interaction state and events capable of manipulating arbitrary 3D content. The platform is intended to support progressively richer sensing and interpretation while preserving stable contracts between layers.

1.4 What the System Is Not

- It is not a conventional AR product whose core value depends on overlaying graphics onto a camera view or wearing a head-mounted display.
- It is not a fixed demo hard-coded around one model, one gesture, or one presentation surface.
- It is not a claim to reproduce fictional MCU capabilities. Tony Stark is an interaction-design inspiration, not an engineering capability baseline.
- It is not a requirement that every future phase replace the previous phase. Later phases augment and refine the platform.

2. System Vision and Engineering Principles

The intended end-state is a real-time spatial interaction platform in which arbitrary 3D content can be inspected and manipulated through natural hand movements, with the interaction quality progressively improved by additional sensing and intelligence. The underlying software should remain independent of any particular content asset and, wherever practical, independent of the physical display technology.

2.1 Core Principles

Principle	Engineering Meaning
Performance first	Smoothness, responsiveness, latency, jitter, stability, and sustained performance are acceptance properties, not cosmetic refinements.
Contract stability	Layer interfaces must permit internal implementation replacement without requiring system-wide rewrites.
Content agnosticism	Interaction semantics operate on generalized scene/object abstractions rather than asset-specific logic.
Display independence	Interaction state must be consumable by different rendering/display backends without semantic changes.

Measurement over perception	Claims of improvement require instrumented evidence and reproducible benchmark conditions.
Graceful degradation	Sensor degradation should reduce confidence or capability predictably rather than silently corrupt state.
Explicit uncertainty	Confidence and temporal validity are part of the data model where relevant.
Evidence-driven evolution	Architectural decisions remain reversible until validated by data, with major choices captured through technical decision records.

2.2 End-State Characteristics

- Natural, low-latency hand interaction suitable for real-time manipulation of arbitrary supported 3D content.
- A unified state model that can accept visual and wearable observations without making either sensor modality the permanent source of truth.
- Reliable interpretation under expected real-world variation, including moderate occlusion, fast movement, lighting variation, and temporary sensor degradation.
- A measurable performance envelope that can be reproduced across controlled benchmark runs.
- An output architecture capable of driving a conventional display today and future spatial-display backends without changing interaction semantics.

3. Development Phases

The project contains exactly three core development phases. Each phase expands system capability while preserving the lower-layer contracts established previously. A phase is complete only when functionality, performance, reliability, testing, documentation, and validation gates are satisfied.

3.1 Phase 1 - Visual Interaction System

Dimension	Specification
Purpose	Establish a high-performance webcam-only interaction baseline on the defined MacBook Air platform.
Input boundary	Built-in webcam video and associated local timing metadata. No additional sensing hardware is assumed.
Output boundary	Interactive 3D scene state presented through the local display backend.
Interaction scope	Selection plus translate, rotate, and scale manipulation; generalized to supported arbitrary 3D objects.
Content scope	Supported .glb, .gltf, and .obj assets, subject to explicit asset validation rules defined by the implementation.
Architecture requirement	The visual pipeline must expose normalized perception/state interfaces so later wearable data can augment it.
Performance posture	Performance-complete. The practical ceiling of smoothness, seamlessness, low latency, low jitter, stable frame delivery, and efficient target-hardware utilization must be characterized and approached.
Hardware boundary	MacBook Air with Apple Silicon M5, 10-core CPU, 10-core GPU, 24 GB unified memory, built-in webcam and display.

Phase 1 Functional Requirements

- P1-F01** The system shall acquire live webcam input and produce a time-ordered visual observation stream.
- P1-F02** The system shall derive hand-related observations sufficient to support the defined interaction set.
- P1-F03** The system shall map hand interaction state into application/scene coordinates without coupling to a single 3D asset.
- P1-F04** The system shall support object selection and manipulation for compatible .glb, .gltf, and .obj content.
- P1-F05** The system shall represent interaction state independently of the rendering backend.
- P1-F06** The system shall handle single-hand and two-hand interaction states with explicit, deterministic state transitions.
- P1-F07** The system shall expose sufficient instrumentation to correlate input, perception, interaction, and rendering timing.

Phase 1 Performance Requirements

- P1-P01** The system shall measure capture, perception/inference, interaction-state update, rendering, and end-to-end latency as separable quantities.
- P1-P02** The system shall measure input FPS, processing FPS, rendering FPS, frame drops, jitter, CPU utilization, GPU utilization, memory use, and sustained thermal behavior.
- P1-P03** The system shall characterize steady-state and sustained-load behavior rather than relying only on short demo runs.
- P1-P04** End-to-end responsiveness shall be optimized toward the practical ceiling of the target hardware, with benchmark evidence documenting the achieved envelope.

P1-P05 The system shall define and track a reproducible baseline so subsequent changes can be evaluated against an established performance reference.

P1-P06 Visual or interaction jitter shall be measurable independently from average throughput, because high FPS does not imply stable interaction.

Phase 1 Exit Gate

- All Phase 1 functional requirements are verified.
- Benchmark instrumentation is operational and produces reproducible measurements.
- The practical performance ceiling of the locked hardware has been characterized under representative workloads and sustained operation.
- Major sources of latency, frame loss, jitter, or instability have been identified and either reduced to an accepted level or explicitly documented as hardware/system limits.
- Single-hand and two-hand behavior has been validated under defined environmental test conditions.
- The Phase 1 architecture exposes the abstraction boundaries required for Phase 2 augmentation without redesigning core interaction semantics.
- Known limitations, benchmark results, and remaining open questions are documented.

3.2 Phase 2 - Wearable Sensor System

Dimension	Specification
Purpose	Introduce a sensorized glove as an additional, higher-rate sensing layer that complements visual observations.
Known planned sensor class	ESP32-class embedded controller, IMU, five flex-sensor channels, contact/touch sensing, haptic vibration outputs, and BLE communication are inherited planning context; exact components remain implementation choices until separately decided.
Input boundary	Wearable telemetry plus device timing/status metadata.
Output boundary	Timestamped, validated wearable observations available to the shared state-estimation pipeline; haptic output is an actuation channel, not a replacement for the visual interaction contract.
Augmentation rule	Wearable sensing augments rather than replaces visual sensing.
Calibration scope	Sensor zeroing, per-user fit/offset handling, cross-sensor alignment, and calibration validity state.
Reliability scope	Dropped packets, stale samples, sensor saturation/faults, communication loss, calibration drift, and power-state transitions shall be represented and observable.
Safety scope	Wearable operation shall consider electrical isolation as applicable, skin/contact safety, heating, mechanical comfort, battery/power safety, and safe failure behavior.

Phase 2 Requirements

P2-F01 The wearable subsystem shall produce timestamped sensor observations with defined channel identity, units, validity, and calibration state.

P2-F02 The subsystem shall provide explicit communication health information sufficient to detect loss, delay, duplication, or staleness.

P2-F03 Wearable observations shall be consumable through a modality-neutral interface rather than being embedded directly into interaction logic.

P2-F04 Calibration state shall be explicit and queryable; interaction logic shall not assume unvalidated sensor data are trustworthy.

P2-F05 The system shall define behavior for temporary wearable unavailability without requiring the visual pipeline to terminate.

P2-F06 Power, thermal, physical-contact, and electrical safety constraints shall be documented and validated before normal-use operation.

P2-P01 Communication and acquisition latency shall be measurable independently from downstream perception and interaction latency.

P2-P02 Sampling-rate stability and timestamp integrity shall be characterized under representative continuous operation.

P2-P03 Wearable data loss and reconnection behavior shall be measurable and tested.

P2-P04 The addition of wearable sensing shall not create an unbounded latency or instability regression in the established Phase 1 path.

Phase 2 Exit Gate

- Wearable observations are timestamped, validated, calibrated, and exposed through stable interfaces.
- Communication, sampling, packet-loss, reconnection, and sensor-fault behavior are quantified.

- Cross-modal data can coexist with the Phase 1 visual stream without requiring visual sensing to be removed.
- Safety and power considerations are documented and tested to the extent applicable to the chosen hardware.
- The system demonstrates stable operation over a sustained representative session.
- The wearable interface is sufficiently abstracted that sensor hardware can later be replaced without redesigning interaction semantics.

3.3 Phase 3 - Sensor Fusion and Intelligent Interaction

Dimension	Specification
Purpose	Combine visual and wearable observations into a unified, temporally consistent interaction state and extend the system with intelligent gesture/intent interpretation.
Fusion boundary	Sensor-agnostic fusion and state-estimation layer consumes modality-specific observations and emits unified hand state plus confidence/validity.
State scope	Hand pose, finger articulation, temporal state, sensor confidence, gesture hypotheses, and interaction intent.
Filtering scope	Temporal smoothing and state estimation are required capabilities; exact algorithms remain an implementation choice to be validated experimentally.
Recognition scope	Gesture recognition shall be separated from low-level sensing and shall expose confidence and temporal context.
Intent scope	Intent interpretation shall map recognized or inferred user actions into stable interaction semantics rather than directly mutating rendering code.
Performance posture	Real-time operation with bounded latency, stable output, and explicit degraded-mode behavior under sensor uncertainty.

Phase 3 Requirements

- P3-F01** The fusion layer shall associate visual and wearable observations using explicit temporal relationships rather than implicit arrival order.
- P3-F02** The unified state shall expose confidence, validity, and temporal freshness for relevant quantities.
- P3-F03** The system shall represent hand pose and finger articulation in a modality-neutral form suitable for downstream interaction logic.
- P3-F04** Gesture recognition shall support explicit confidence and temporal stability rather than emitting unqualified instantaneous labels.
- P3-F05** Intent interpretation shall be decoupled from content-specific object logic.
- P3-F06** The interaction engine shall remain usable when one modality becomes degraded, subject to defined capability reduction.
- P3-P01** The fused pipeline shall expose stage-level and end-to-end latency metrics comparable with earlier phase benchmarks.
- P3-P02** Fusion shall preserve deterministic timestamp semantics and provide measurable handling of stale, missing, or conflicting observations.
- P3-P03** Filtering and state estimation shall be evaluated for both latency and stability; smoothing that materially increases responsiveness latency shall not be accepted solely because it looks visually cleaner.
- P3-P04** Recognition quality shall be evaluated quantitatively using defined test sets and error metrics appropriate to the gesture/intent domain.
- P3-P05** The system shall maintain a predictable degraded mode when confidence falls below defined operational thresholds.

Phase 3 Exit Gate

- Visual and wearable streams are time-associated and represented through stable contracts.
- Unified state, confidence, and degradation behavior are verified.
- Gesture/intent recognition demonstrates quantified reliability on the defined evaluation set.
- Latency, jitter, throughput, and resource utilization remain within the accepted system envelope.
- Sensor failure and degraded-operation tests demonstrate safe, predictable behavior.
- All material architectural decisions and experimental conclusions are recorded.

4. System Architecture

The architecture is organized as a layered pipeline. The layers define responsibilities and contracts, not a mandatory process topology or specific software implementation. Implementations may combine or split runtime components provided the externally visible responsibilities and interfaces remain stable.

Layer	Responsibility Boundary
Sensor / Input Layer	Produces modality-specific raw observations. Owns device-facing concerns and source metadata. Must not encode object manipulation semantics.
Acquisition Layer	Captures, timestamps, validates, buffers, and transports observations. Owns sequencing and source health, not interpretation.
Perception Layer	Transforms raw inputs into structured observations such as visual landmarks or wearable articulation channels. Exposes confidence and validity.
State-Estimation Layer	Associates observations over time, derives a unified hand state, handles temporal continuity, and represents uncertainty.
Gesture / Intent Layer	Interprets state trajectories into gestures and user intent hypotheses. Must remain independent of individual assets and display mechanics.
Interaction Layer	Converts intent into stable interaction commands and object-independent interaction events.
Content / Scene Layer	Owns 3D scene state, object transforms, selection state, hierarchy, and asset-independent scene semantics.
Rendering / Display Layer	Consumes scene state and presents it through a selected display backend. Must not be the source of interaction truth.
Instrumentation / Observability Layer	Collects timing, counters, resource, quality, error, calibration, and benchmark metadata across all layers.

4.1 Boundary Rules

- Perception shall not directly manipulate scene objects.
- The interaction layer shall not depend on a particular renderer.
- The scene layer shall not depend on how hand observations were generated.
- Sensor-specific packet formats shall terminate at the acquisition/perception boundary rather than propagate into scene logic.
- Instrumentation shall observe runtime behavior without becoming a hidden source of timing semantics or correctness state.
- Any cross-layer shortcut must be treated as an explicit architectural exception and recorded as a technical decision when material.

5. Data Contracts and Interface Semantics

Interfaces are conceptual contracts. Exact serialization, class structures, schemas, and programming-language representations are intentionally left open. Every time-varying observation must carry enough metadata to establish provenance, timing, validity, and interpretation.

5.1 Canonical Data Domains

Data Domain	Minimum Semantics
Raw sensor observation	Source identity, timestamp, sequence/reference information where applicable, raw value(s), units or encoding, validity/status.
Visual observation	Camera/source identity, timestamp, image/frame reference or derived observation reference, detection/tracking status, confidence, image-space geometry.
Hand landmarks	Hand identity, ordered landmarks, coordinate space, timestamp, confidence/visibility per relevant quantity, tracking status.
Hand pose	Position/orientation representation, coordinate space, timestamp, validity, confidence, and temporal continuity metadata.
Finger articulation	Per-finger articulation quantities, units, source provenance, timestamp, confidence, calibration state.
Temporal state	Time index, state age/freshness, predicted/interpolated/observed status where relevant, continuity state.
Gesture hypothesis	Gesture identifier, phase/state if applicable, confidence, onset/continuation/termination semantics, supporting temporal interval.
Interaction event	Semantic action, target abstraction, parameters, timestamp, source state reference, confidence/authorization state.
3D object state	Object identity, transform, selection state, hierarchy/context, visibility, and content metadata required for rendering.

5.2 Contract Properties

DC-01 Every time-varying observation shall have an explicit timestamp or time reference sufficient for temporal association.

DC-02 Coordinates shall identify their coordinate space explicitly; implicit coordinate assumptions are prohibited across component boundaries.

DC-03 Units shall be explicit for physical quantities and stable within each contract.

DC-04 Validity and confidence shall be distinguishable. Invalid data are not equivalent to low-confidence but usable data.

DC-05 Data consumers shall tolerate additional fields or metadata where schema evolution requires it, subject to versioning rules.

DC-06 Interaction events shall describe semantic intent/action, not renderer-specific instructions.

DC-07 Object state shall remain independent from the source modality that caused its modification.

6. Coordinate Systems and Timing

6.1 Coordinate Spaces

Space	Definition / Requirement
Camera coordinates	Coordinates relative to the camera/sensing frame. Camera orientation, handedness, units, and origin must be explicit.
Image coordinates	2D coordinates tied to image/frame geometry. Pixel dimensions, orientation, mirroring, and normalization conventions must be explicit.
Hand/local coordinates	Coordinates anchored to a hand or hand-relative reference frame. Local origin and orientation convention must be defined.
World/application coordinates	Stable application-space representation used for interaction and scene reasoning.
Object coordinates	Local coordinates of an individual 3D asset prior to application/world transforms.
Transform relationships	Mappings between spaces must be explicit, versioned when semantics change, and testable independently of gesture logic.

6.2 Timing Contract

TIME-01 Each source shall expose a coherent timestamp domain and document its time origin and resolution.

TIME-02 Cross-source association shall be based on timestamps or another explicitly defined temporal correspondence mechanism, not incidental queue arrival order.

TIME-03 The system shall distinguish capture time, processing time, presentation time, and observation-to-action time when measuring latency.

TIME-04 Sampling frequency shall be measurable for each real-time stream; average rate alone is insufficient where burstiness matters.

TIME-05 Frame/observation association shall be traceable sufficiently to diagnose drops, duplication, reordering, or stale data.

TIME-06 Latency budgets shall be monitored as stage contributions so that optimization can target the actual limiting stage.

7. Performance Engineering

Performance is a continuous engineering requirement. The system shall treat throughput, responsiveness, latency, jitter, and stability as distinct properties.

7.1 Metrics

Metric	Required Interpretation
Input FPS / sampling rate	Rate at which source observations are actually acquired, including variability.
Processing FPS / throughput	Rate at which observations are processed end-to-end through a defined stage.
Rendering FPS	Rate at which presentation frames are generated/displayed, measured rather than inferred.
Frame drops	Missing, skipped, duplicated, or late frames/observations relative to the expected stream.
End-to-end latency	Elapsed time from the relevant physical/input observation to the corresponding rendered/presented effect, with the measurement definition documented.
Stage latency	Time consumed by acquisition, perception, estimation, recognition, interaction, rendering, and other material stages.
Jitter	Variation of a timing or spatial quantity over time, reported using an explicitly defined statistic and test window.
CPU utilization	Processor demand attributable to the application/system under a reproducible workload.
GPU utilization	Graphics/accelerator demand under the same workload and observation window.
Memory usage	Resident/unified memory demand, peak usage, and growth across sustained sessions.
Thermal behavior	Performance stability and any sustained-load degradation over defined session durations.

7.2 Performance Methodology

- Every benchmark shall document hardware state, application configuration, content workload, environment, duration, and measurement definitions.
- Median values alone are insufficient for real-time systems; distributions or percentile statistics should be retained where useful for latency and jitter analysis.
- A benchmark change shall state both improvement and regression effects. Optimizing one stage at the expense of another is not considered an unconditional improvement.
- Short transient peaks and long-duration sustained behavior shall be evaluated separately.
- Results shall be reproducible enough to determine whether a claimed change exceeds ordinary measurement noise.

7.3 Phase 1 Performance Priority

HARDWARE LOCK: Phase 1 performance conclusions are specific to the MacBook Air M5 configuration defined in this document. External cameras, depth sensors, wearable hardware, or other compute devices are not baseline Phase 1 assumptions.

- The practical smoothness ceiling shall be measured rather than declared.
- End-to-end latency shall be reduced as far as the chosen architecture and locked hardware permit without compromising required correctness and stability.
- Landmark/hand-state jitter shall be measured independently from visual frame rate.
- Thermal and sustained behavior shall be included in the exit assessment because the platform is fanless and short benchmark runs can misrepresent long-session performance.

8. Testing and Validation

8.1 Test Domains

Domain	Required Coverage
Functional correctness	Selection, translation, rotation, scaling, state transitions, multi-hand behavior, asset loading, scene consistency.
Performance	Throughput, latency, jitter, resource use, frame loss, sustained behavior.
Tracking stability	Stationary hand, slow motion, rapid motion, transitions, recovery from temporary loss.
Gesture reliability	Confusable gestures, onset/termination, repeated execution, fatigue/variation, confidence behavior.
Environmental robustness	Lighting variation, background contrast, motion blur, clutter, camera positioning within supported setup.
Occlusion	Partial hand occlusion, object occlusion, self-occlusion, temporary sensor blockage.
Multiple hands	One-hand baseline, two-hand interaction, inter-hand ambiguity, hand identity continuity.
Sensor degradation	Dropped frames/packets, stale data, communication loss, calibration invalidity, low-confidence perception.
Long-duration operation	Memory growth, thermal behavior, resource drift, calibration drift, connectivity stability.

8.2 Validation Levels

- Component validation: individual layer behavior and contract semantics.
- Integration validation: boundary behavior between acquisition, perception, state estimation, interaction, content, and rendering.
- System validation: complete real-time workflows under benchmark and stress conditions.
- Regression validation: repeatable benchmark suite before/after meaningful architecture or performance changes.
- Field-style validation: representative use conditions outside ideal laboratory lighting or motion assumptions.

8.3 Quantitative Acceptance

Acceptance thresholds should be established where empirical evidence can define meaningful limits. Thresholds must distinguish hard safety/correctness limits from optimization targets. A target may evolve as hardware limits become understood; a safety or data-integrity constraint may not be weakened merely for convenience.

9. Calibration and Reliability

9.1 Calibration

Calibration Area	Requirements
Camera-based sensing	Camera/image geometry assumptions must be known and validated for the supported configuration. Any runtime transformations must be reproducible.
Wearable sensors	Zero/reference state and per-channel calibration validity shall be represented. Sensor drift and recalibration behavior shall be measurable.
Cross-sensor alignment	Relationships between wearable and visual representations shall have an explicit calibration contract and validation procedure.
User-specific calibration	Where fit, hand size, sensor placement, or range changes behavior, the calibration state shall be explicit and recoverable.
Runtime confidence	The system shall surface confidence/validity changes caused by degraded sensing, calibration invalidation, or environmental uncertainty.

9.2 Reliability and Degradation

REL-01 A missing or degraded sensor shall not silently produce apparently valid data.

REL-02 Consumers shall be able to determine whether state is current, stale, predicted, degraded, or invalid where such distinctions exist.

REL-03 Temporary sensor loss shall have a defined recovery behavior and shall be observable in diagnostics.

REL-04 When a modality becomes unreliable, the system shall degrade capability in a bounded and predictable manner.

REL-05 The system shall avoid abrupt destructive state changes caused solely by transient sensor noise or a single missing sample, unless the interaction contract explicitly requires immediate termination.

10. Modularity and Extensibility

The architecture shall treat sensing and presentation technologies as replaceable modules. The primary stable asset of the system is its semantic interaction pipeline, not any one hardware or rendering implementation.

Extension Target	Required Abstraction
Cameras	Source-neutral observation contract with explicit geometry/timing metadata.
Wearable sensors	Device-neutral telemetry contract with calibration and health metadata.
Tracking systems	Perception interface producing normalized observations independent of the tracker implementation.
Gesture models	Model/input-independent gesture hypothesis contract with confidence and temporal semantics.
3D assets	Asset-neutral scene/object contract with import validation separated from interaction semantics.
Rendering engines	Scene/state consumption interface independent of presentation technology.
Display technologies	Output backend interface accepting scene/presentation state without altering interaction semantics.

MOD-01 Phase 1 visual sensing assumptions shall not become hard dependencies of Phase 2 or Phase 3 semantic interfaces.

MOD-02 Phase 2 wearable packet semantics shall not leak into the gesture/intent or scene layers.

MOD-03 Phase 3 fusion shall consume modality abstractions and shall not make a specific sensor model structurally mandatory unless that dependency is documented and justified.

MOD-04 Display backends shall consume scene/presentation state rather than owning gesture or intent interpretation.

MOD-05 Replacing one perception implementation with another shall not require rewriting content/scene semantics.

11. Future Spatial Display Compatibility

Future display technologies are compatibility targets, not current implementation dependencies. The core architecture shall preserve a clean distinction between interaction state and its presentation.

- Conventional displays
- Projection-based systems
- Multi-view systems
- Light-field systems
- Volumetric systems

- AR/spatial displays
 - DISP-01** Interaction events and scene state shall remain meaningful independent of display topology.
 - DISP-02** Presentation transforms or display-specific rendering adaptations shall terminate at the rendering/display boundary.
 - DISP-03** A future display backend shall not require the sensor or gesture layer to know its optical or spatial characteristics except through an explicit capability contract where necessary.
 - DISP-04** The project shall not make AR headset tracking, headset optics, or headset availability a prerequisite for core system validity.

12. Security, Safety, and Privacy

12.1 Camera and Sensor Data

- Prefer local processing for real-time perception where practical within the project architecture.
- Document whether raw camera frames, derived landmarks, wearable telemetry, and benchmark recordings are retained.
- Avoid retaining raw sensor data when derived data are sufficient for an experiment, unless raw data are explicitly required for reproducibility or debugging.
- Protect stored datasets and configuration artifacts from accidental exposure or unauthorized modification.

12.2 Wearable and Hardware Safety

- Wearable power systems shall operate within the safe electrical and thermal characteristics of selected components and enclosure/garment integration.
- Physical contact surfaces shall be evaluated for irritation, pressure, heat, sharp edges, and entanglement risks.
- Any external connectivity path shall have a defined trust boundary and failure behavior.
- High-voltage, plasma, mains, or unisolated experimental hardware is outside the project safety boundary for future demonstrator work.

12.3 Connectivity

- SEC-01** External connectivity shall not be required for core real-time interaction unless a future explicitly approved feature depends on it.
- SEC-02** Network-connected components shall expose only the interfaces necessary for their function and shall define authentication or trust assumptions where applicable.

13. Observability and Experimentation

The system shall be designed so engineering claims can be supported by evidence. Instrumentation is part of the product architecture, not a disposable debugging aid.

Capability	Requirement
Logging	Structured logs for material lifecycle events, faults, calibration changes, state transitions, and experiment metadata.
Metrics	Continuous capture of latency, throughput, frame loss, jitter, quality, resource, and health metrics appropriate to each phase.
Profiling	Ability to identify material time and resource consumers at stage level.
Benchmark datasets	Versioned, documented representative conditions for performance and recognition evaluation.
Configuration tracking	Experiment runs must record the configuration needed to explain or reproduce materially different results.
Comparative evaluation	Changes should be compared against a named baseline using the same benchmark definitions whenever possible.
Traceability	Where practical, a user-visible interaction result should be traceable back to the relevant observation/state path for diagnosis.

13.1 Experiment Integrity

- OBS-01** A performance result without a documented measurement definition is not considered a valid engineering baseline.
- OBS-02** Benchmark configuration changes shall be recorded rather than silently changing the meaning of historical results.
- OBS-03** Tests intended to compare two implementations shall control or document environmental differences that can materially affect the result.

OBS-04 Optimization claims shall report relevant regressions or trade-offs when a change moves another quality attribute.

14. Engineering Constraints, Assumptions, Dependencies, and Non-Goals

14.1 Current Constraints

- Phase 1 hardware is locked to the specified MacBook Air M5 and built-in webcam/display.
- Phase 1 does not assume external cameras, depth sensors, wearable devices, or external accelerators.
- The system must support arbitrary compatible 3D content rather than a single fixed asset.
- Phase completion is gated by evidence, not by feature count.
- The core roadmap contains exactly three phases.

14.2 Phase-Specific Constraints

- Phase 1: webcam-only sensing.
- Phase 2: wearable sensing is additive and must coexist with the visual stream.
- Phase 3: multimodal fusion and intelligent interaction are the primary algorithmic expansion; exact models/filters remain implementation choices.

14.3 Future Assumptions

- Future display technologies may exist with different optical, geometric, and rendering constraints.
- Future sensing devices may provide different sampling rates, accuracy, or coordinate conventions.
- Future model and inference choices may change as benchmark evidence accumulates.

14.4 Non-Goals

- Reproducing MCU-fiction capabilities as literal engineering functionality.
- Making AR headsets a prerequisite.
- Freezing specific implementation libraries, language choices, or algorithms before validation requires them.
- Building the deferred separate energy/arc-reactor demonstrator as part of this core project.

14.5 Unresolved Dependencies

- Actual webcam performance envelope under varying lighting and motion conditions.
- Sustained thermal behavior and any associated performance drift on the fanless target hardware.
- Empirical limits of tracking accuracy, motion blur tolerance, and multi-hand stability.
- Validated communication and power behavior of the eventual wearable design.

15. Formal Phase Gates

Gate	Entry Criteria	Exit Criteria
Phase 1	Architecture baseline established; benchmark definitions written; target hardware confirmed.	Functional requirements verified; performance instrumentation operational; practical hardware performance ceiling characterized; representative robustness tests passed; documentation and baseline evidence complete; Phase 2 interfaces structurally preserved.
Phase 2	Phase 1 exit accepted; wearable design contracts reviewed; safety requirements defined.	Wearable telemetry reliable and measurable; calibration/state contracts valid; communication and fault behavior quantified; coexistence with visual sensing demonstrated; safety and sustained operation validated.
Phase 3	Phase 2 exit accepted; unified data contracts and fusion evaluation plan reviewed.	Cross-modal synchronization, unified state, confidence, gesture/intent reliability, degraded modes, performance, and full validation evidence accepted; decision records and unresolved questions updated.

GATE PRINCIPLE: Primary functionality working is necessary but never sufficient. Every gate requires functional correctness plus performance, reliability, validation, observability, and documentation evidence.

16. Technical Decision Records

Material architectural choices shall be captured as Technical Decision Records (TDRs). This section defines the record format; it does not fabricate decisions that have not yet been made.

Field	Required Content
Decision	The chosen architectural or engineering decision, stated precisely.
Context	The problem, constraint, evidence, and conditions that made a decision necessary.
Alternatives considered	Material alternatives evaluated or deliberately not evaluated, with reasons.
Trade-offs	Benefits, costs, performance effects, complexity, reversibility, and risks.
Rationale	Evidence-based explanation for the selected option.
Consequences	Expected architectural, operational, maintenance, and future-phase implications.
Status	Proposed, accepted, superseded, deprecated, or rejected, with date/version as applicable.

- A TDR should be created before a major irreversible dependency is introduced.
- A TDR may remain provisional while an experiment is underway.
- Superseding a TDR must reference the earlier decision and explain what evidence changed the conclusion.
- Implementation detail that does not affect architectural contracts need not become a TDR.

17. Open Engineering Questions

The following are deliberately unresolved. Future implementation or research work shall answer them using measurement, experimentation, or documented technical evidence. The specification does not silently convert these unknowns into decisions.

17.1 Phase 1 Hardware and Capture

What is the maximum stable capture rate and practical exposure/motion-blur envelope of the built-in webcam under controlled and representative lighting conditions?

17.2 Phase 1 Thermal Behavior

What sustained CPU/GPU/resource load can the fanless MacBook Air maintain before measurable degradation occurs, and how does that alter the real-time performance envelope?

17.3 Phase 1 Perception Ceiling

What tracking quality and temporal stability can be achieved under fast motion, self-occlusion, background clutter, and two-hand interaction?

17.4 Coordinate Mapping

What coordinate representation and calibration contract most reliably maps webcam observations into stable application-space interaction without introducing unacceptable latency or jitter?

17.5 Phase 2 Wearable Design

Which concrete sensor components and physical arrangement provide the required fidelity, comfort, power behavior, and data rate without unnecessary complexity?

17.6 Communication

What telemetry transport characteristics provide sufficient reliability and latency for the wearable while preserving clear failure detection and recovery semantics?

17.7 Cross-Sensor Calibration

How should visual and wearable coordinate systems be aligned and revalidated after sensor movement, glove refit, or user change?

17.8 Fusion Strategy

Which state-estimation and filtering approaches provide the best measured trade-off among responsiveness, smoothness, robustness, and compute cost?

17.9 Recognition

Which gesture and intent representations generalize across users and environmental variation while keeping false activation acceptably low?

17.10 Degraded Modes

What confidence thresholds and fallback behaviors produce the safest and least surprising interaction when modalities disagree or temporarily fail?

17.11 Display Independence

What minimum presentation capability contract allows future projection, multi-view, light-field, volumetric, or AR/spatial displays to consume the same interaction/scene state?

17.12 Benchmark Governance

What benchmark corpus, scenario set, statistical treatment, and run protocol are sufficient to distinguish genuine improvements from noise or environment-specific gains?

18. Specification Change Control

- CHG-01** Changes to architectural requirements, phase boundaries, core data contracts, or phase gates shall be explicitly recorded with revision history.
- CHG-02** A change shall identify affected requirements, interfaces, tests, and prior decision records where relevant.
- CHG-03** New implementation discoveries shall not silently redefine the canonical contract; proposed changes must be reviewed against existing constraints and future-phase modularity.
- CHG-04** Superseded requirements shall remain traceable so historical implementation decisions can be understood.

18.1 Revision History

Revision	Status	Summary
1.0	Baseline	Canonical three-phase engineering specification established from current project constraints and handover context.

Appendix A. Requirement Identifier Scheme

Requirement identifiers use phase or domain prefixes to support implementation traceability. Examples: P1-F for Phase 1 functional, P1-P for Phase 1 performance, P2-F/P2-P for Phase 2, P3-F/P3-P for Phase 3, DC for data contracts, TIME for timing, REL for reliability, MOD for modularity, DISP for display compatibility, SEC for security, OBS for observability, and CHG for change control.

End of canonical technical specification.